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B.Tech. Degree VI Semester Regular Examination April 2022

ME 19-205-0607 ENERGY CONSERVATION AND ENVIRONMENT PROTECTION
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Gain a basic understanding and appreciation of energy efficiency and environmental concepts.

CO2: Learn basic operating principles of day to day energy conservation devices.

CO3: Discuss various options to increase energy efficiency.

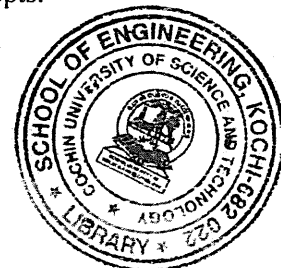
CO4: Examine ways to save energy and money.

CO5: Explore ways to save the environment.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze,

L5 – Evaluate, L6 – Create

PO – Programme Outcome



PART A

(Answer *ALL* questions)

I.	(8 × 3 = 24)	Marks	BL	CO	PO
(a) List the strategies for better energy security of the nation.	3	3	L2	2	1,2
(b) List down the advantages of 'simple payback period' technique.	3	3	L2	1	1,2
(c) What is energy auditing? Discuss its general scope.	3	3	L1	1	1
(d) Enumerate the methods of testing steam traps.	3	3	L2	2	1,2
(e) Write short note on the environmental effects of nuclear power generation.	3	3	L1	1	1,2
(f) What are the implications of Global warming?	3	3	L2	1	1,2
(g) Explain the term Air Quality Index.	3	3	L1	1	1
(h) Discuss the impacts of water pollution.	3	3	L2	2	1,2

PART B

(4 × 12 = 48)

- II. (a) Give a detailed account of fossil fuel usage pattern among the world countries and give the limitations of fossil fuel. 4 L4 3 2,3
- (b) Calculate the NPV of two projects and suggest which of the two projects should be accepted assuming discount rate of 10%. 8 L5 4 3,4

	Project A	Project B
Initial Investment (₹)	40,000	60,000
Estimated life	5 years	5 years
Scrap Value (₹)	2,000	4,000

Cash inflows					
Year	1	2	3	4	5
Project A	10,000	20,000	20,000	6,000	4,000
Project B	40,000	20,000	10,000	6,000	4,000

OR

(P.T.O)

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- III. (a) Explain the trends in energy use patterns. 4 L4 3 1,2,3
- (b) Calculate the Internal Rate of Return of an investment of ₹1,36,000 which yields the following cash inflows. 8 L5 4 3,4

Year	Cash Inflows (₹)
1	30,000
2	40,000
3	60,000
4	30,000
5	20,000

- IV. Explain the ten steps methodology for the detailed energy auditing. 12 L4 4 3,4
- OR**
- V. (a) Distinguish between 'preliminary energy audit' and 'detailed energy audit'. 6 L4 3 3,4
- (b) Explain condensate recovery system in a steam system. 6 L4 4 3,4
- VI. Write the cogeneration systems based on: 12 L3 3 3,4
- (i) Diesel Engine
- (ii) Gas Turbine.
- OR**
- VII. Write short notes on: 12 L3 4 3,4
- (i) Geothermal energy
- (ii) Tidal energy.
- VIII. (a) Distinguish between primary and secondary air pollutants. 4 L4 5 4,5
- (b) What is the scope of environmental auditing in a country like India and explain the challenges? 8 L5 4 4,5
- OR**
- IX. (a) Explain the types of air pollutants and their impacts. 4 L4 5 4,5
- (b) Explain the detailed procedure of exhaust emission test in a vehicle. 8 L3 5 5

Blooms's Taxonomy Levels

L1-10%, L2-12.5%, L3-26.6%, L4-30%, L5-20%, L6-0%
